

2026 May Term (ALU)

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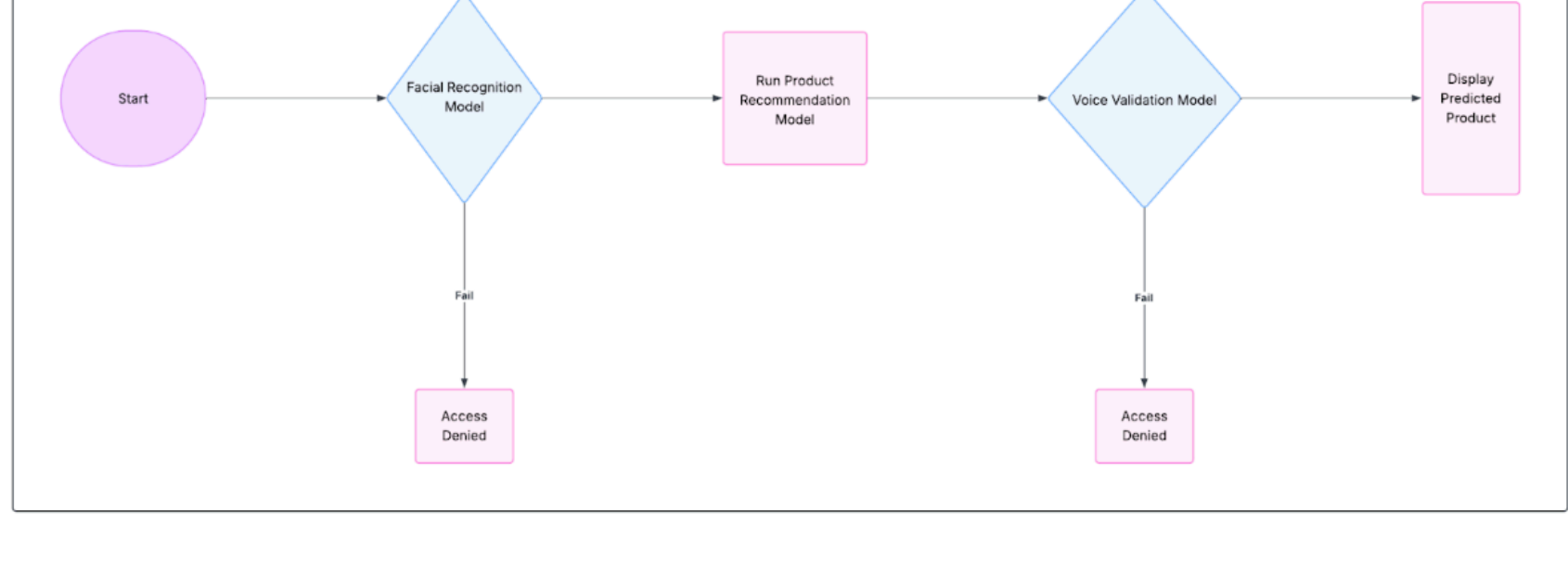
Start Assignment

Due Sunday by 11:59pm Points 40 Submitting a file upload
File Types pdf Attempts 0 Allowed Attempts 2
Available Jul 8 at 12am - Jul 24 at 11:59pm

Formative 2: Multimodal Data Preprocessing Assignment

Assignment Overview

This diagram outlines the User Identity and Product Recommendation System Flow, a sequential process designed to authenticate a user through facial recognition and voice validation before providing personalized product recommendations. The system incorporates multiple security checkpoints, with clear access denied pathways if any authentication step fails, ensuring that product recommendations are only displayed to verified users.



The flow:

1. A user attempts to access the product prediction model.
2. If recognized, they proceed to attempt to run a prediction.
3. The prediction must be confirmed through voice verification.
4. The system uses pre-trained models to determine whether:
 1. The face matches a known user (face recognition)
 2. The voice is an approved voice sample (voiceprint verification)
 3. The prediction is allowed

Assignment Instructions

- Merge Tabular Data and Create a Prediction Model
- Collect and preprocess image and audio data
- Apply image-based and audio-based authentication methods
- Merge and engineer multimodal features for ML use
- Train and evaluate a model that simulates real-world decision logic

Tasks

1. Data Merge

You are provided with two datasets [customer_social_profiles](#) and [customer_transactions](#) . The goal is to create a predictive model using a dataset that is created as a result of merging the two sources. The model predicts the product a customer would purchase when visiting your social media sites. You are tasked with ensuring you capture the relevant features from both datasets.

2. Image Data Collection and Processing

Each group member must:

- Submit at least 3 facial images: one neutral, one smiling, and one surprised.
- Load and display sample pictures of each member
- Apply augmentations per image (e.g., rotation, flipping, grayscale).
- Extract and save image features (e.g., embeddings, histograms) into image_features.csv.

3. Sound Data Collection and Processing

- Record at least 2 audio samples saying phrases like:

- “Yes, approve”

- “Confirm transaction”

- Load and display sound samples as waveforms and spectrograms for each member
- Apply at least two augmentations per sample (e.g., pitch shift, time stretch, background noise).
- Extract and save features (e.g., MFCCs, spectral roll-off, energy) into audio_features.csv

4. Model Creation

Develop the following models using your engineered and merged dataset:

- Facial Recognition Model
- Voiceprint Verification Model
- Product Recommendation Model

Use models such as: Random Forest or Logistic Regression or XGBoost, etc

Evaluate performance using metrics such as: Accuracy, F1-Score and Loss

6. System Demonstration

- Simulate at least one image and one audio sample for "unauthorized attempt"
- Simulate a full transaction:
 - Input face image → Allows you to make a call to the product model
 - Input voice → approves & displays the user to execute the prediction
- Implement as a script or mini-app on the command line

Deliverables

You will submit a report describing your approach in detail, a system simulation video link, the GitHub repository link, and every team member's contribution. Make sure to have the following details in your GitHub repository:

- Datasets
 - Merged dataset with feature engineering
 - image_features.csv and audio_features.csv
- Python Scripts for the pipeline (for all three models) used to implement the command-line app
- Jupyter notebook

Formative 2 - Data Preprocessing						
Criteria	Ratings				Pts	
Exploratory Data Analysis Quality & Insight	4 to >3.0 pts Exemplary Summary statistics, variable types, ≥3 labeled plots (dist, outliers, correlations)	3 to >2.0 pts Proficient Basic stats and ≈2 plots with minimal interpretation	2 to >1.0 pts Developing Only 1 plot or basic stats; limited insight	1 to >0 pts Beginning No meaningful EDA or interpretation	4 pts	
Data Cleaning & Merge Validation	4 to >3.0 pts Exemplary Nulls/duplicates handled, types fixed; join logic justified; post-merge checks shown	3 to >2.0 pts Proficient Some cleaning, basic join, validation incomplete	2 to >1.0 pts Developing Inconsistent merge, minimal cleaning, no justification	1 to >0 pts Beginning Merge failed or skipped; no data prep	4 pts	
Image Quantity & Diversity	4 to >3.0 pts Exemplary Each member: 3 expressions (neutral, smile, surprised); consistent naming and format	3 to >2.0 pts Proficient All members submit the required images; minor issues	2 to >1.0 pts Developing Incomplete expressions or images missing	1 to >0 pts Beginning Few usable images; missing team data	4 pts	
Image Augmentation & Feature Extraction	4 to >3.0 pts Exemplary ≥2 augmentations per image; embeddings/histograms saved correctly in image_features.csv	3 to >2.0 pts Proficient 1 augmentation; basic features extracted	2 to >1.0 pts Developing Some augmentations; poor or unclear feature file	1 to >0 pts Beginning No augmentation or meaningful features	4 pts	
Audio Quality & Visualization	4 to >3.0 pts Exemplary 2 phrases per member; clean audio; waveform and spectrogram plotted and interpreted	3 to >2.0 pts Proficient Phrases present for all; visualizations shown	2 to >1.0 pts Developing Some members missing audio or visuals	1 to >0 pts Beginning No audio or unusable recordings	4 pts	
Audio Augmentation & Feature Extraction	4 to >3.0 pts Exemplary ≥2 augmentations; MFCCs + roll-off/energy saved in audio_features.csv	3 to >2.0 pts Proficient 1 augmentation; features extracted	2 to >1.0 pts Developing Weak augmentations; minimal or messy features	1 to >0 pts Beginning No processing or saved features	4 pts	
Model Implementation	4 to >3.0 pts Exemplary All 3 models (face, voice, product) implemented, trained, and functional	3 to >2.0 pts Proficient All models present but minor issues in logic or flow	2 to >1.0 pts Developing 1–2 models working; poor flow or bugs	1 to >0 pts Beginning Models broken or missing	4 pts	
Evaluation & Multimodal Logic	4 to >3.0 pts Exemplary Each model evaluated (accuracy, F1, loss); multimodal logic explained and used	3 to >2.0 pts Proficient At least one metric per model; limited logic	2 to >1.0 pts Developing Metrics are weak or logic is not clearly multimodal	1 to >0 pts Beginning No metrics or logic shown	4 pts	
System Simulation	4 to >3.0 pts Exemplary Full transaction + unauthorized demo working in CLI/Jupyter; smooth interaction	3 to >2.0 pts Proficient Valid simulation works; unauthorized attempt buggy	2 to >1.0 pts Developing Partial/hardcoded flow; weak interaction	1 to >0 pts Beginning No working simulation	4 pts	
Submission Quality & Communication	4 to >3.0 pts Exemplary Report, notebook, code, and files are clean, named, and well-documented with concrete details	3 to >2.0 pts Proficient Minor clarity issues; all items present	2 to >1.0 pts Developing Messy structure; weak explanations or missing items	1 to >0 pts Beginning Poorly organized or incomplete	4 pts	
Total Points: 40						

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